

GUDIPATOJU NITESH

AI/ML Engineer | Generative AI & LLMs | Python & Backend Engineering

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SUMMARY

AI/ML Engineer (B.Tech AI & ML) specializing in Generative AI and applied machine learning with Python. Built a JWT-secured RAG chatbot (FastAPI, ChromaDB, Docker) and tuned an XGBoost crop-advisory model to ~90% accuracy using GridSearchCV and 5-fold cross-validation. Currently applying computer vision and deep learning as an AI Intern at XTRAGRAD, and extending into agentic AI with JobPilot (FastAPI, LangChain/LangGraph).

TECHNICAL SKILLS

Languages: Python, Java, SQL

Machine Learning: Scikit-learn, XGBoost, Random Forest, Gradient Boosting, GridSearchCV, Cross-Validation, Feature Engineering, Model Evaluation

Generative AI: RAG (Retrieval-Augmented Generation), ChromaDB, LLM Integration (Llama 3.3 70B via Groq), LangChain/LangGraph, Prompt Engineering, Vector Embeddings, Agentic AI, Whisper

Computer Vision: Deep Learning, Real-Time Detection, Webcam-Based Monitoring

Backend & Deployment: FastAPI, Flask, REST APIs, JWT Authentication, PostgreSQL, SQLAlchemy, Docker, Git/GitHub

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn

EXPERIENCE

AI Intern, XTRAGRAD Pvt. Ltd. (AICTE Internship Program)

Sep 2026 – Present

- Selected for a structured AI/ML internship focused on guided assignments and project-based learning under technical mentorship.
- Building an AI-based Online Exam Monitoring System using computer vision and deep learning to detect predefined abnormal activities from webcam video during online exams, with real-time alerts.
- Developing a Flask web application with a monitoring dashboard to surface live alerts and exam session activity.

Data Analytics Virtual Experience (Simulation), Deloitte / Forge

Sep – Oct 2025

- Performed forensic-style financial data analysis on a simulated client dataset using Python and Excel; built ETL scripts to automate reporting.

AI/ML Data Analytics Job Simulation, Tata Group / Forge

Sep 2025

- Conducted exploratory data analysis and applied AI-based prediction models to a business risk-profiling case study.

PROJECTS

AI Chatbot with Personality [\[GitHub\]](#)

Nov 2025 – Jan 2026

LLM, RAG, ChromaDB, Whisper STT, FastAPI, Docker, JWT

- Designed a ChromaDB-based RAG pipeline, benchmarking retrieval quality against a keyword-based baseline; built and secured 5 REST endpoints with FastAPI and JWT, containerized with Docker.
- Integrated OpenAI Whisper for speech-to-text and engineered persona-driven prompts to sustain coherent personality across multi-turn conversations.

AI Smart Irrigation System with Weather-Aware Crop Guidance [\[GitHub\]](#)

Jan – Mar 2026

Python, XGBoost, Random Forest, GridSearchCV, 5-Fold CV, PostgreSQL, FastAPI

- Trained and tuned an XGBoost crop-prediction model on 10,000+ sensor readings using GridSearchCV and 5-fold cross-validation, reaching ~90% accuracy — outperforming a Random Forest baseline by ~15%; findings presented as an IEEE conference contribution.
- Built a weather-aware crop-advisory module on FastAPI and PostgreSQL that ingests live weather data to generate irrigation guidance.

JobPilot — Agentic AI Job Search & Application Assistant [\[GitHub\]](#)

In Development

Python, FastAPI, Groq API, Llama 3.3 70B, PostgreSQL, SQLAlchemy, LangChain/LangGraph

- Architected the backend foundation for an agentic job-search assistant powered by Groq-hosted Llama 3.3 70B with PostgreSQL/SQLAlchemy persistence.
- Developing agent orchestration, tool-calling, and a human-approval workflow with LangChain/LangGraph.

Student Placement Prediction Dashboard [\[GitHub\]](#)

Aug – Dec 2025

Python, Random Forest, Scikit-learn, Streamlit, 5-Fold CV

- Trained a 5-fold cross-validated Random Forest classifier on 2,500+ student records, reaching ~82% placement-prediction accuracy; deployed as an interactive Streamlit dashboard.

EDUCATION

B.Tech, Artificial Intelligence & Machine Learning

2022 – 2026

Nalla Malla Reddy Engineering College, Hyderabad | CGPA: 8.0 / 10

CERTIFICATIONS

- Introduction to Generative AI Studio — Google Cloud (Aug 2025)
- Generative Models for Developers — Infosys Springboard (Mar 2026)

ADDITIONAL

- AI Tryouts — Independent AI & technology content platform, 34K+ followers; plans, creates, and publishes content on AI, education, and freelancing, driving audience growth and engagement.
- Software Testing Intern — EduRankAI, National Internship Portal (2026).